Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

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Abstract

Low-resource regional dialects pose severe challenges for Natural Language Processing (NLP) systems due to significant morpho-syntactic variation, non-standard orthography, and extreme scarcity of annotated parallel corpora. In this paper, we address the task of dialect normalization—converting non-standard regional dialectal text into Standard Written Marathi—focusing on three major Indo-Aryan regional dialects of Maharashtra: Southern Konkan, Northern Konkan, and Varhadi. To overcome severe data scarcity, we introduce an automated synthetic parallel data expansion pipeline combining large language model (LLM) generation using Gemma-4 (12B) with a multi-tier verification engine (rule-based script filtering and LLM semantic auditing), effectively doubling the clean parallel training corpus into verified synthetically expanded pairs across agriculture and finance domains. We conduct a rigorous empirical evaluation of state-of-the-art multilingual sequence-to-sequence transformer architectures—google/mT5-small (300M) and IndicBART (244M)—under an 85/15 stratified test split with 5-fold cross-validation across 8 dataset configurations. Empirical results demonstrate that mT5-small achieves state-of-the-art performance with 69.51 BLEU and 84.58 chrF++ on the synthetically expanded multi-dialect dataset (+12.39 BLEU over IndicBART), with individual dialect performance reaching 81.00 BLEU and 91.21 chrF++ on Varhadi. We present detailed architectural comparisons, data scaling ablation studies, and qualitative subword morphological error analyses, establishing a benchmark for Indic dialect text normalization.

Keywords: Dialect Normalization , Low-Resource NLP , Marathi Dialects , Sequence-to-Sequence Modeling , Synthetic Data Augmentation , mT5 , IndicBART , Text Standardization.

Introduction

Rapid development of deep learning architectures and self-attention mechanisms [1] has made it possible to develop systems capable of understanding and generating natural language. Open-weight models are increasingly deployed to process text across various contextual domains, as they allow finer control over model architecture, runtime environment, and training data [2]. Even with these advancements, current language systems predominantly cater to standard languages, yielding substandard outputs when queried with non-standard regional dialects [3]. Speakers of regional dialects routinely encounter severe performance degradation across automated speech recognition, machine translation, and text processing systems due to distinct lexical, phonetic, and morpho-syntactic variations [4–6].

Dialect normalization is the task of transducing non-standard regional dialectal text into standard written language while preserving original sentence semantics [5]. It represents an essential layer for digital linguistic inclusion. In linguistically diverse nations such as India—home to 22 officially recognized languages and hundreds of regional dialects spoken by 1.4

billion people—sub-regional varieties remain severely under-represented in computational research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India [7]. Because its regional dialects possess distinct lexical, phonetic, and morpho-syntactic properties, we hypothesize that model performance and evaluation metrics will vary significantly across different dialectal varieties.

Building pipelines for translating Indic dialects primarily comes down to two core challenges:

1. **Extreme Data Scarcity**: Datasets that map regional dialects to standard forms are technically non-existent; and
2. **Morphological Complexity**: Marathi is a language with rich morphology, where noun cases, postpositional affixes, and verbal tense suffixes exhibit complex variations [7, 8].

Addressing these limitations, this work presents a multi-dialect Marathi text normalization framework combining LLM-driven synthetic data augmentation with sequence-to-sequence transformer fine-tuning. We undertake this study because the performance of existing multilingual models on regional Marathi dialects remains largely unevaluated, and because automated synthetic data expansion—when constrained by multi-tier verification—can reliably enhance output accuracy for severely resource-constrained tasks.

Our core contributions are:

- **Parallel Corpus Generation**: Based on transcripts from the RESPIN dataset [9], we construct an aligned parallel corpus for three major Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi across agricultural and banking domains.
- **Automated Synthetic Augmentation with Verification**: We design a synthetic data expansion pipeline using Gemma-4 [10] paired with a multi-tier verification engine (rule-based script filtering and LLM semantic auditing), demonstrating that audited synthetic data significantly improves downstream model accuracy.
- **Cross-Dialect Evaluation & Model Benchmarking**: We empirically demonstrate that normalization performance varies substantially across dialects, and establish comprehensive benchmarks for existing multilingual models (mT5-small vs. IndicBART) across single-dialect and multi-dialect training configurations.

Related Work

Regional Dialect Standardization across Language Families

Early work in European dialect normalization relied on Character-Level Statistical Machine Translation (CSMT) and phrase-based SMT for Swiss German Dialects [11], which were subsequently superseded by pre-trained language models and neural sequence-to-sequence architectures [12, 13]. Parallel efforts have addressed dialectal normalization across Uralic and Scandinavian varieties, including Finnish, Swedish, and Estonian [14–16]. A systematic multilingual evaluation across four language families by Kuparinen et al. [17] demonstrated that fine-tuned sequence-to-sequence transformers consistently outperform rule-based and statistical baselines. Beyond European languages, East Asian and Afro-Asiatic dialect processing has evolved along similar lines: Japanese regional dialects have been modeled via phrase-level LSTMs [18], Mandarin non-standard varieties via acoustic-phonetic mapping [19], and Dialectal Arabic via the Conventional Orthography for Dialectal Arabic (CODA) framework [20], enabling multi-city translation benchmarks (MADAR [21], NADI [22]) and morphology-aware prompting strategies [23, 24].

Indic Dialect Landscape and Marathi NLP

While pre-trained multilingual language models such as IndicBART [25], IndicNLP Suite [26], and mT5 [27] cover major Indic national languages, they suffer from significant performance degradation on unscripted regional sub-dialects. Monolingual suites like L3Cube-MahaNLP [28] provide pre-trained Marathi models for classification and tagging, but lack sequence transduction capabilities for generative text standardization. In the broader Indic context, INDIC-DIALECT [29] introduced translation benchmarks for Hindi variants (Bhojpuri, Magahi, Maithili) evaluated using Weighted Finite-State Transducers (WFST) [30], while Dhasmana et al. [31] explored cross-lingual transfer for low-resource Indic varieties. However, research on Marathi regional dialects has remained largely restricted to speech/ASR classification [32, 33], zero-shot retrieval heuristics [34, 35], or preliminary rule-based text mapping [36]. Our work addresses this gap by presenting an end-to-end neural fine-tuning pipeline supported by multi-tier synthetic data augmentation and rigorous cross-dialect benchmarks.

LLM Distillation and Multi-Tier Verification

Large Language Models (LLMs) are increasingly leveraged to alleviate data scarcity across generation tasks [37–39]. Knowledge distillation and synthetic data expansion enable the training of compact, efficient task-specific models using outputs generated by larger teacher architectures [40–42]. However, applying unconstrained synthetic generation to low-resource regional dialects introduces substantial risks of hallucination, non-standard orthographic noise, and language leakage [43]. We mitigate these risks by combining Gemma-4 generation with a morphologically tailored prompt design and a closed-loop, multi-tier verification engine to guarantee dataset integrity prior to neural fine-tuning.

Methodology

Target Dialects and Linguistic Properties

The following Marathi dialects from the RESPIN corpus were selected for this study, as they exhibit unique lexical and phonetic variations:

1. **Southern Konkan**: Shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms are its key characteristics. Lexical differences include vowel shifts: standard जातो (“he goes”) becomes जातां; standard कुठे (“where”) becomes कुठंय.
2. **Northern Konkan**: Since it is surrounded by neighboring Gujarati and Bhili language families, it tends to exhibit significant consonant variations. Few examples are: standard कशाला (“why”) $\rightarrow$ Northern Konkan कसाले; standard मुलगा (“boy”) $\rightarrow$ Northern Konkan पोर.
3. **Varhadi**: Spoken in the regions of Vidarbha accompanied with verb suffix modifications. Standard questioning का? (“why?”) becomes काऊन?; verbal forms exhibit noticeable patterns. Varhadi maintains closest structural similarity to Standard Marathi as compared to other dialects since the transformations are more systematic.

Original Corpus Curation and Foundation in RESPIN

Our primary data source is the RESPIN-S1.0 initiative, developed by researchers at IISc Bangalore. RESPIN provides audio recordings and their transcripts in agriculture and banking domains for different Indian languages and their dialects. However, the corpus contains only raw dialect transcripts—it does not provide aligned dialect-to-standard Marathi translations, which are essential for training normalization models.

To bridge this gap, we used Gemma-4 (12B configuration, accessed via Ollama, run locally) to translate each dialect transcript into Standard Marathi, thereby constructing the original parallel corpus. The unique prompts from each dialect partition (5,838 for Southern Konkan, 5,825 for Northern Konkan, and 5,330 for Varhadi) were individually processed through the LLM translation pipeline described in Section 3.3, yielding 16,163 verified parallel pairs across three dialects.

LLM-Driven Translation and Verification Pipeline

Both the original parallel corpus construction (Phase 1) and the subsequent synthetic expansion (Phase 2) rely on the same LLM-driven translation and verification pipeline. In Phase 1, each RESPIN dialect transcript is translated into Standard Marathi to create the initial parallel pairs. In Phase 2, additional synthetic dialect–standard pairs are generated from the same

Table 1: RESPIN-S1.0 Marathi Source Corpus Statistics

Code	Dialect Name	Utterances (%)	Duration (h)	Speakers	Unique Prompts
D1	Southern Konkan (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkan (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	**All Dialects**	**809,934 (100%)**	**1,026.06**	**2,644**	**23,069**

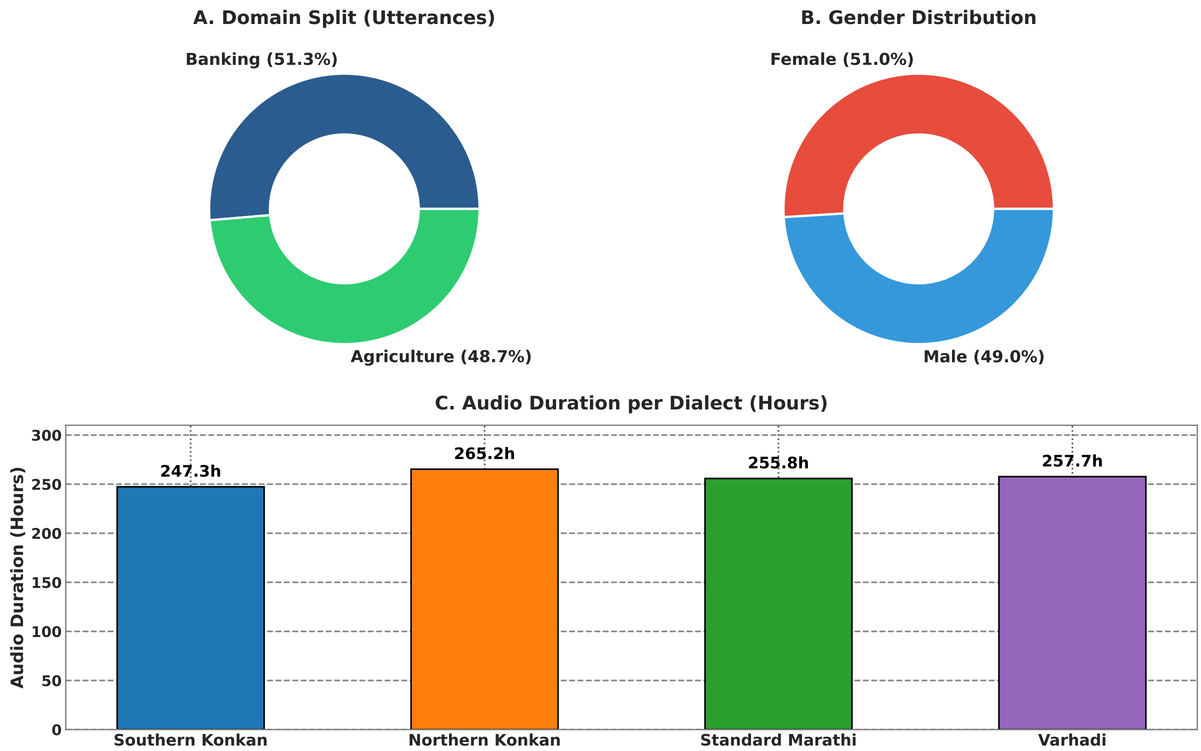

Figure 1: RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution; (B) gender split; (C) audio recording duration per dialect variant.

source transcripts to expand the training corpus. The pipeline employs the following design aspects:

- **1-Prompt-Per-Sentence Architecture**: Instead of batching multiple pairs per API call (which risks contextual loss), each sentence is processed separately.
- **Detailed Prompting Strategy**: System prompt is designed for the output to preserve exact meaning, context regarding the domain, proper nouns, and numerical values from the input with a set of 5-8 few-shot examples per dialect.
- **Temperature Control**: Generation temperature is set to 0.3 with top-p = 0.9 for optimal results.

All translations are generated using Gemma-4 under the 12B configuration accessed via Ollama, run locally. Every generated pair—whether from Phase 1 or Phase 2—undergoes a three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.**
    - *Devanagari Script Integrity*: Both source and target must have at least 90% Devanagari Unicode characters that lie in the `U+0900--U+097F` range, that automatically get rid of unsolicited characters.
    - *Length Ratio Constraint*: $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$ to avoid hallucinations.
    - *Non-Identity Constraint*: $L_{\text{dialect}} \neq L_{\text{standard}}$ to flag identical string copies.
    - *Minimum Token Count*: Both source and target must contain at least 3 whitespace-

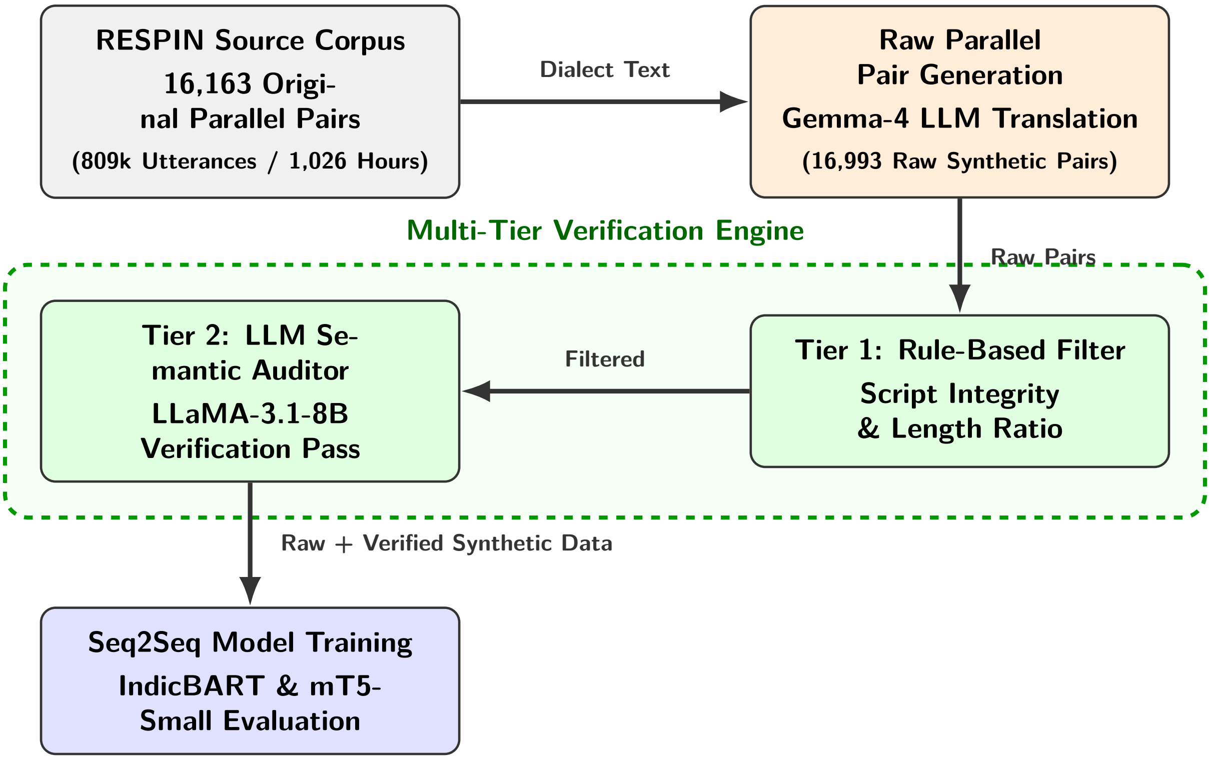

Figure 2: System architecture of the multi-dialect normalization framework.

Table 2: Statistics based on generated and verified data

Dialect Name	Original Clean Pairs	Raw Synthetic Generated	Clean Synthetic Verified	Corrupted / Rejected Pairs (%)	Total Clean Expanded Pairs
Southern Konkan	5,576	5,838	5,569	269 (4.61%)	**11,145**
Northern Konkan	5,501	5,825	5,534	291 (5.00%)	**11,035**
Varhadi	5,086	5,330	5,069	261 (4.90%)	**10,155**
Total	**16,163**	**16,993**	**16,172**	**821 (4.83%)**	**32,335**

separated tokens.

2. **Tier 2: LLM Semantic Verification.** A secondary round of evaluation was performed using `llama-3.1-8b-instant` (accessed via Groq API). The verification happens on a scale of 1–5 score that considers semantics, hallucinations, multilingual contamination, and complete standardization of dialects. Pairs that score < 4.0 are rejected or routed for further correction.
3. **Closed-Loop Correction Engine.** Pairs that are flagged by Tier 1 or Tier 2 enter a correction loop: Step 1 generates corrected translations using specialised prompts based on the pre-defined reason of identified failure (e.g., "Hindi leakage detected" or "Incomplete normalization"); Step 2 verifies the corrected sentences again, following the complete discard of pairs that fail the second audit pass.

Synthetic Data Expansion (Phase 2)

After constructing the original 16,163 verified parallel pairs through Phase 1, we applied the same pipeline to synthetically expand the training corpus. The expansion phase re-processes the source dialect transcripts under varied prompting conditions to generate additional dialect–standard pairs, effectively doubling the training data.

The pipeline was used to generate parallel pairs from the RESPIN transcripts and for synthetic data expansion. Out of the 16,993 synthetic parallel pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 flawed pairs (an overall corruption/rejection rate of 4.83%), yielding 16,172 clean verified pairs as mentioned in Table 2. Combining with the 16,163 clean original pairs, the final expanded dataset consists of 32,335 verified

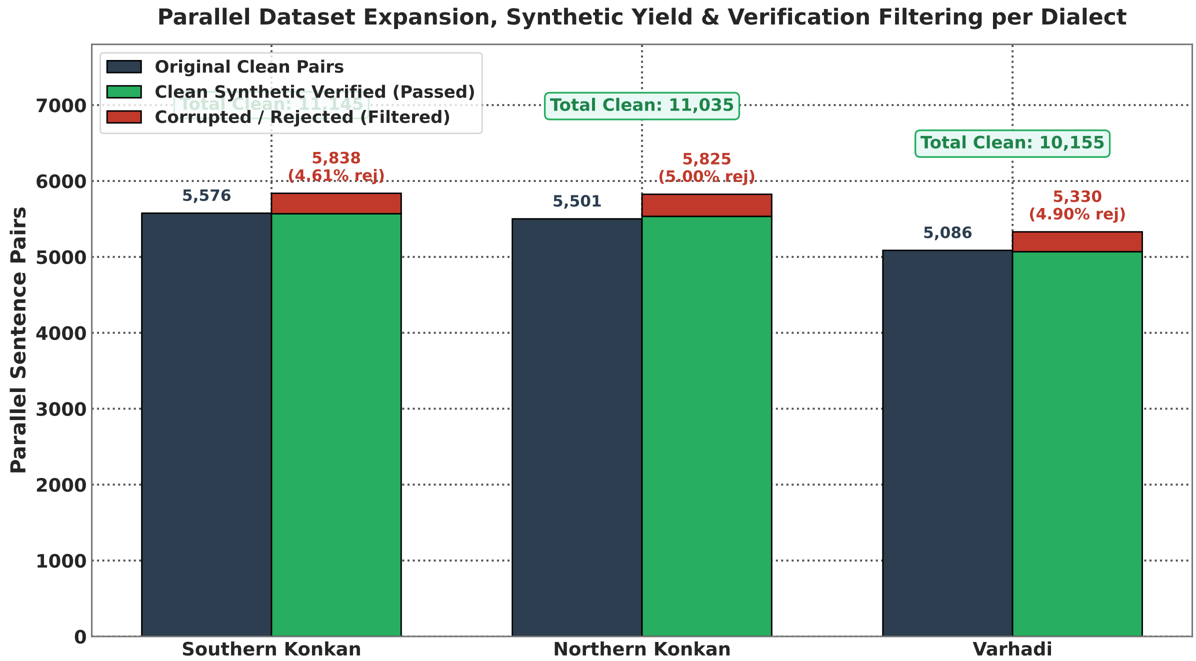

Figure 3: Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

parallel sentence pairs across all three dialects.

Evaluation Protocol

For stable performance during development, models were validated using 5-Fold Cross-Validation on an 85/15 stratified split of the train set. The final reports were generated on the official held-out test set provided by RESPIN after training the model on the complete dataset, which had around 2,170 utterances across target dialects. This setup is applied across all dialects on the original transcript-based data and their synthetically expanded versions. Results are reported using metrics that are evaluated using SacreBLEU [44] and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- **chrF++**: Character n-gram F-score combined with word unigrams and bigrams.
- **Normalized WER / CER**: Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

Findings

Comprehensive Test Benchmark on `IISc_RESPIN_test_mr`

Key observations from Table 3 and Figure 4 include:

1. **mT5-Small Performance**: `mT5-small` upon finetuning on the synthetically expanded dataset achieves 73.48 BLEU and 87.70 chrF++ on the overall test set.
2. **IndicBART Scaling**: Fine-tuning IndicBART on the synthetically expanded multi-dialect corpus achieves **76.50 BLEU** and **88.04 chrF++**, suggesting that larger data size brings up the efficiency in mBART-50 architectures.

Single-Dialect vs. **Multi-Dialect** Benchmark Results on Original and Synthetic Datasets

Table 4 presents the performance matrix on original single-dialect benchmarks, and Table 5 explains the dialect-wise breakdown of the multi-dialect model on original data.

Table 6 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 7 details the dialectwise evaluation breakdown of the multi-dialect model on

Table 3: Comprehensive Evaluation Matrix on `IISc_RESPIN_test_mr` Benchmark Set (2,170 Utterances)

Model Scope	Training Data	Dialect / Subset	Utts	IndicBART (244M)		mT5-Small (300M)	
				BLEU (↑)	WER (↓)	BLEU (↑)	WER (↓)
Single-Dialect	Original	Southern Konkan	559	**57.80**	**26.60%**	43.86	34.37%
		Northern Konkan	540	**90.13**	**6.46%**	79.46	11.95%
		Varhadi	516	**83.59**	**10.19%**	74.81	14.73%
	Synthetically Expanded	Southern Konkan	559	24.06	60.32%	**44.36**	**32.75%**
		Northern Konkan	540	65.41	28.70%	**79.76**	**12.61%**
		Varhadi	516	67.97	25.43%	**76.90**	**14.19%**
Multi-Dialect	Original	Southern Konkan	559	26.21	53.15%	**40.94**	**35.34%**
		Northern Konkan	540	47.59	32.84%	**77.50**	**14.65%**
		Standard Benchmark	555	96.12	3.12%	**96.65**	**1.91%**
		Varhadi	516	72.75	25.30%	**73.47**	**16.16%**
	Synthetically Expanded	Southern Konkan	559	**52.06**	36.80%	43.88	**34.96%**
		Northern Konkan	540	49.08	28.70%	**78.16**	**13.55%**
		Standard Benchmark	555	96.80	1.65%	**97.39**	**1.37%**
		Varhadi	516	70.27	21.40%	**74.74**	**15.57%**

Table 4: Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	5,576	837 / 836	**48.54**	**78.35**	46.31	74.43
Northern Konkan (D2)	5,501	826 / 825	**60.58**	**80.30**	60.31	77.34
Varhadi (D4)	5,086	763 / 762	76.63	90.93	**81.00**	**91.21**

expanded data.

Impact of Synthetic Data Augmentation

Table 8 isolates the effect of dataset scaling by comparing performance across original and synthetically expanded corpora for each model architecture. Overall, multi-dialect models exhibit significant net gains (+8.95 BLEU for IndicBART, +6.22 BLEU for mT5-small), confirming that data scarcity was a primary performance bottleneck.

But a dialect-wise breakdown suggests notable variance across the corpora. Southern Konkan showed significant gains from expansion (+25.85 BLEU for IndicBART, +18.34 BLEU for mT5-small), while Northern Konkan sees moderate gains (+1.49 BLEU for IndicBART, +0.37 BLEU for mT5-small) due to its already high original baseline performance. We attribute this behavior to two factors:

1. **Baseline Saturation and Precision Dilution**: Varhadi already had better performance on original non-expanded set (72.75-79.57 BLEU). We estimate that LLM-generated synthetic pairs may introduce suffix rules which were over-generalized that slightly degraded the exact N-gram matching against the original reference texts.
2. **Cross-Dialect Parameter Competition**: In shared dialect training, we estimate that appending synthetic data across distinct sub-dialects may have caused performance competition in shared attention weights, which could have considered prioritizing recovery on severely sparse dialects (Southern Konkan) over saturated ones.

Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

Performance of our verification and filtering engine is measured by conducting a comparative analysis on models trained on *Raw Unverified Synthetic Data* versus *Filtered Clean Syn-*

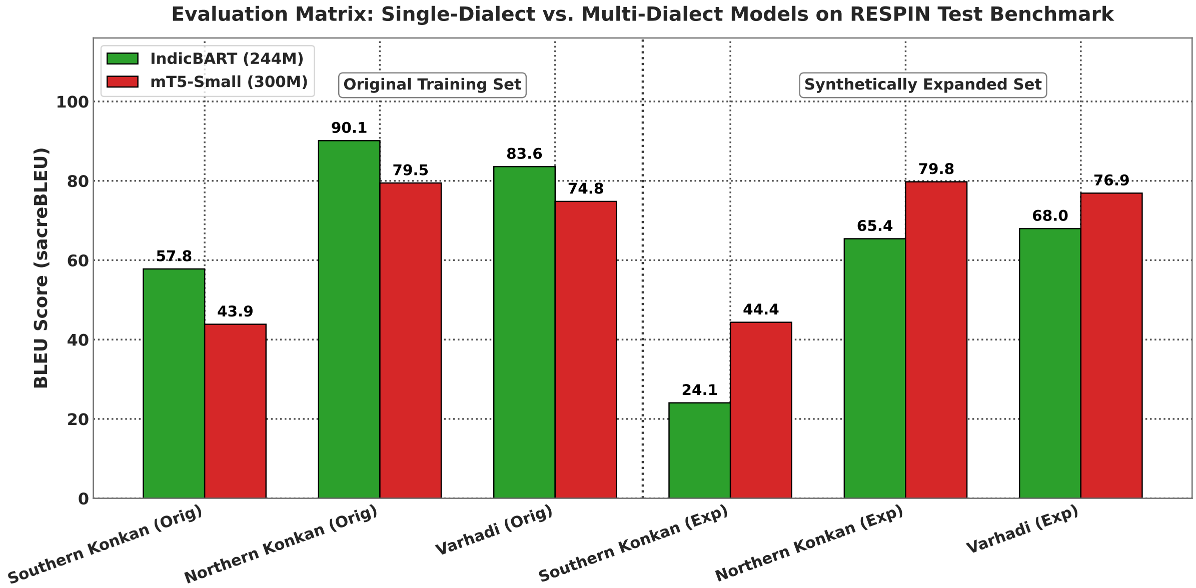

Figure 4: Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

Table 5: Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	16,163	837 / 836	26.21	53.15	**48.28**	**75.04**
Northern Konkan (D2)	16,163	826 / 798	47.59	67.16	**62.35**	**78.92**
Varhadi (D4)	16,163	763 / 790	72.75	85.63	**79.57**	**90.51**
Overall Combined	**16,163**	**2,426 / 2,424**	**48.17**	**68.58**	**63.29**	**81.37**

thetic Data (the expanded suite). The raw unverified dataset contains 19,914 parallel pairs, that included 3,742 rejected pairs that contained garbage translations. Table 9 presents the quantitative comparison on the benchmark suite.

Key findings and insights from this evaluation include:

1. **Quality Collapse on Unverified Data**: Training IndicBART on unverified outputs causes a significant drop of 14.03 BLEU points, collapsing from 57.12 to 43.09 due to compounded error propagation from uncorrected dialect residue and wrong syntax with other errors.
2. **mT5-Small Resilience**: While google/mT5-small showed resilience due to its 250,000-token multilingual capacity, training on unverified data still suffers an 8.30 BLEU point drop, falling from 69.51 to 61.21 and a 4.40 chrF++ drop.
3. **Essential Role of Multi-Tier Auditing**: Removal of flawed ones and correcting the possible set of pairs is directly responsible for better performance that eliminated subsequent errors.

Model Architecture Analysis

Table 10 provides a detailed architectural comparison between the two primary models. The 69.51 vs. 57.12 BLEU gap on the expanded multi-dialect task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage**: mT5's 250K vocabulary provides 3.9$\times$ more subword units than

Table 6: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	11,145	1,672 / 1,671	47.25	64.69	**65.10**	**82.88**
Northern Konkan (D2)	11,035	1,656 / 1,655	40.51	62.21	**62.07**	**79.29**
Varhadi (D4)	10,155	1,524 / 1,523	73.62	86.75	**78.89**	**90.59**

Table 7: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialect-wise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	32,335	1,672 / 1,710	52.06	72.15	**66.62**	**83.57**
Northern Konkan (D2)	32,335	1,656 / 1,627	49.08	69.90	**62.72**	**79.63**
Varhadi (D4)	32,335	1,524 / 1,513	70.27	84.27	**78.73**	**90.46**
Overall Combined	**32,335**	**4,852 / 4,850**	**57.12**	**75.49**	**69.51**	**84.58**

IndicBART's 64K, enabling finer-grained tokenization of morphologically complex Marathi dialectal forms. This is critical for dialect normalization, where subtle suffix and infix changes must be captured at the subword level.

2. **Positional Encoding**: mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead**: IndicBART requires explicit language tokens (`<2mr>`) that add conditioning complexity, whereas mT5's pure text-to-text format enables direct mapping without task-specific architectural constraints.

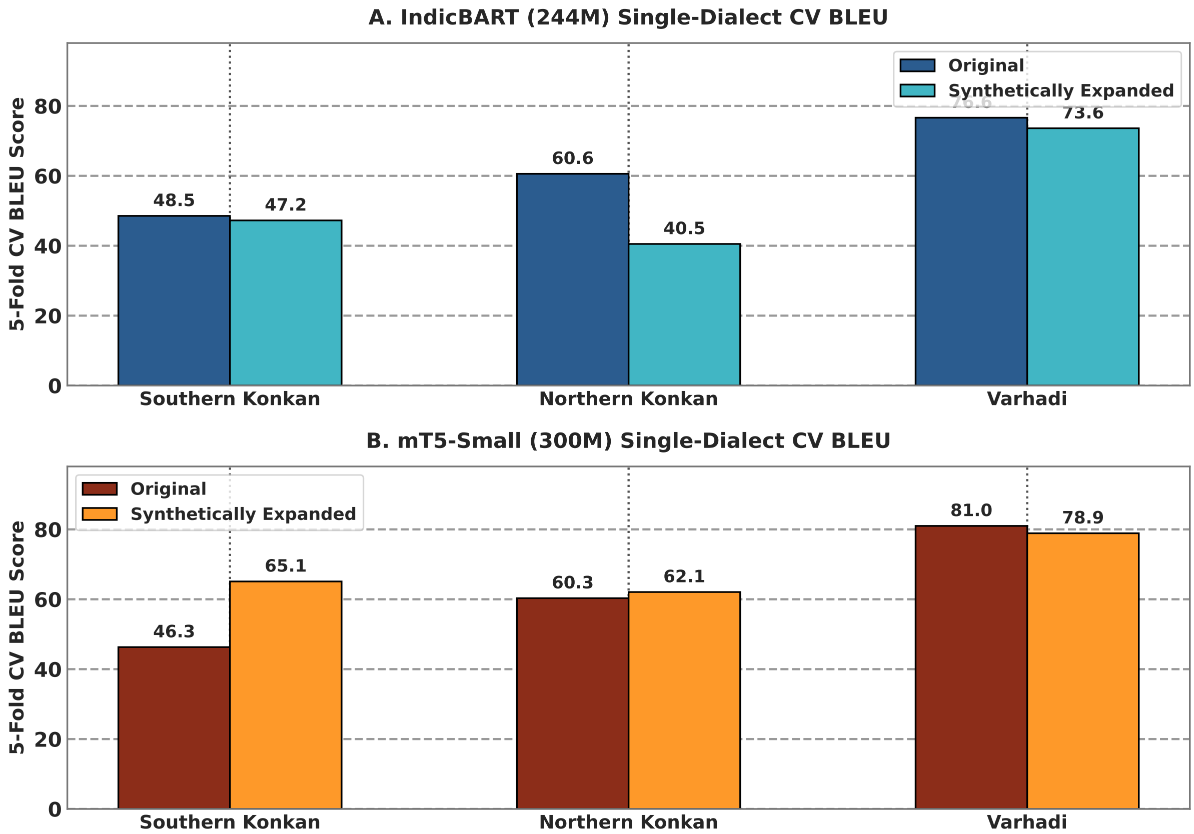

Figure 5: Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

Table 8: Impact of Synthetic Data Augmentation: Original vs. Synthetically Expanded Data BLEU Change (5-Fold CV)

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkan (D1)	26.21	52.06	**+25.85**	48.28	**66.62**	**+18.34**
Northern Konkan (D2)	47.59	49.08	**+1.49**	62.35	**62.72**	**+0.37**
Varhadi (D4)	**72.75**	70.27	−2.48	**79.57**	78.73	−0.84
Overall Combined	**48.17**	**57.12**	**+8.95**	**63.29**	**69.51**	**+6.22**

Table 9: Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

Model	Pipeline State	Total Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	−14.03	−10.95
	Filtered Clean Data	32,335	**57.12**	**75.49**	**+14.03**	**+10.95**
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	−8.30	−4.40
	Filtered Clean Data	32,335	**69.51**	**84.58**	**+8.30**	**+4.40**

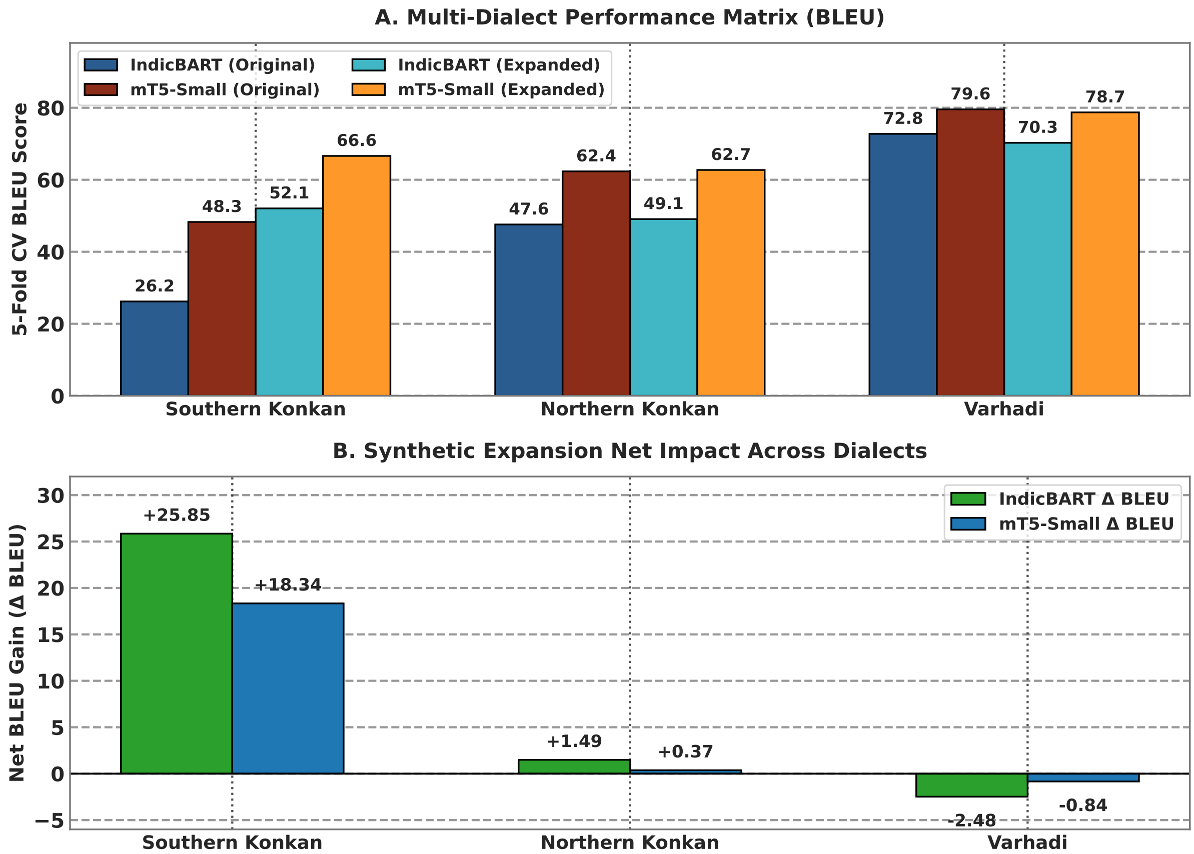

Figure 6: Multi-dialect 5-fold cross-validation benchmark comparison: (A) Absolute BLEU scores across model architectures and corpora partitions, and (B) Net synthetic expansion impact (Δ BLEU across sub-dialects).

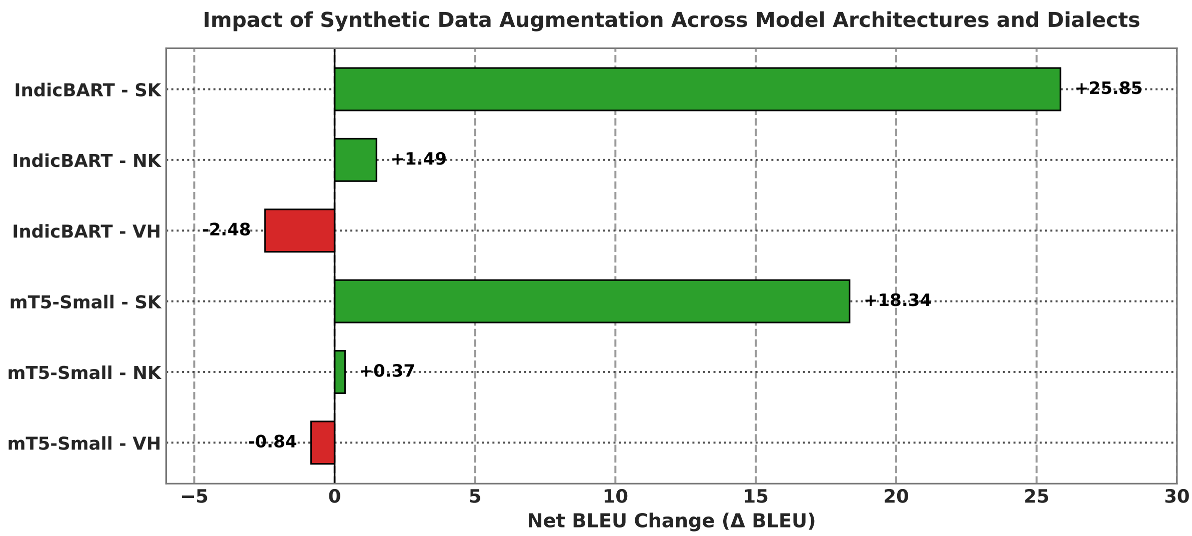

Figure 7: Impact of LLM synthetic data expansion (Δ BLEU score comparing synthetically expanded vs. original 5-fold cross-validation benchmarks).

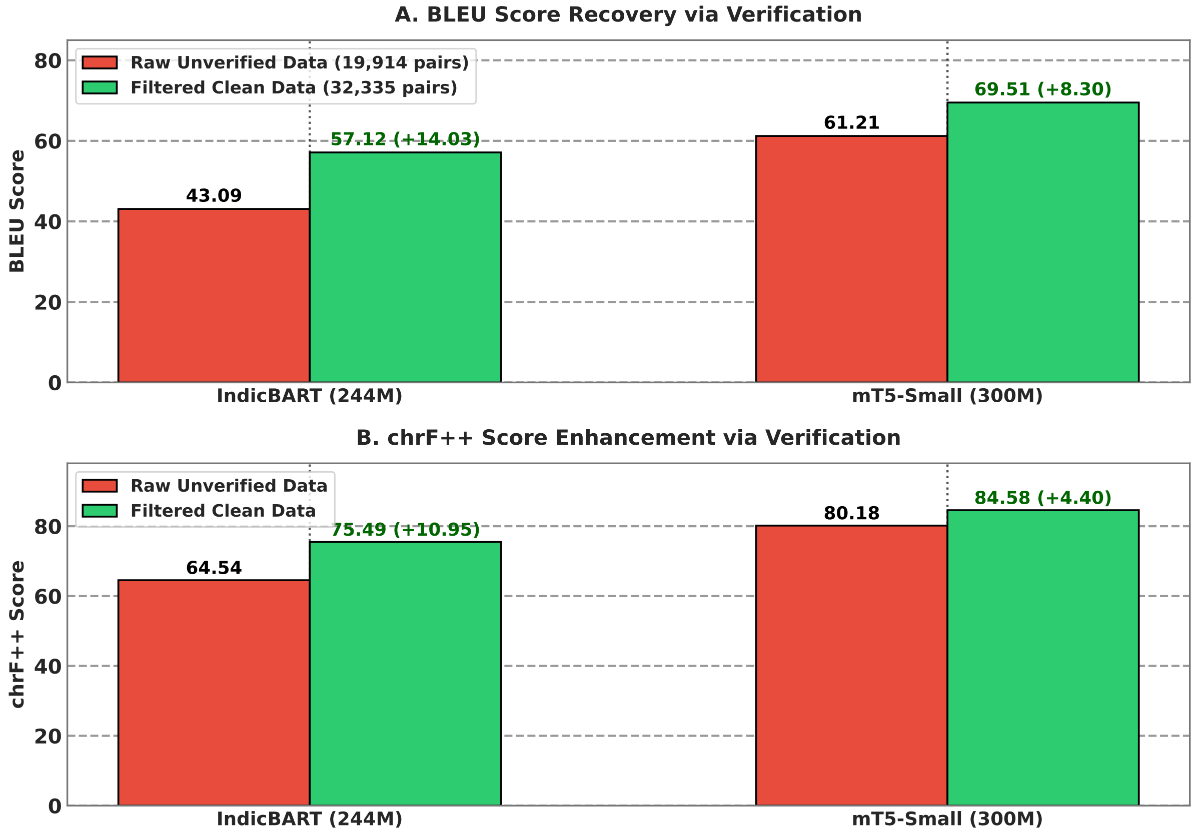

Figure 8: Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

Table 10: Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	**IndicBART (244M)**	**mT5-small (300M)**
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required `<2mr>` token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

Conclusion

This study presents a comprehensive, reproducible framework for multi-dialect Marathi text normalization, investigating dialectal variance under normalization, the efficacy of verified synthetic augmentation, and the performance of pre-trained multilingual sequence-to-sequence models on low-resource Indic dialects.

Our empirical findings confirm that normalization difficulty varies significantly across regional dialects: individual dialect performance ranged from 66.62 BLEU (Southern Konkan) to 78.73 BLEU (Varhadi) for mT5-small on the expanded multi-dialect benchmark, highlighting the necessity of dialect-specific evaluation rather than treating Marathi as a monolithic target.

Our LLM-driven synthetic data augmentation pipeline, powered by Gemma-4 with multi-tier verification (Devanagari script filtering and LLaMA-3.1-8B semantic auditing), successfully expanded the clean parallel corpus from 16,163 to 32,335 pairs while maintaining high semantic fidelity. Multi-tier verification proved crucial for augmentation quality, yielding a recovery of +14.03 BLEU for IndicBART and +8.30 BLEU for mT5-small compared to training on raw, unverified synthetic outputs.

Evaluating existing pre-trained multilingual architectures on the expanded benchmark established new state-of-the-art baselines: google/mT5-small (300M) achieved 69.51 BLEU and 84.58 chrF++ on the multi-dialect dataset, outperforming IndicBART (244M) by +12.39 BLEU. This performance margin is largely driven by mT5's 3.9$\times$ larger subword vocabulary (250K vs. 64K tokens) and relative positional biases, which accommodate the morphological complexity of dialectal Marathi more effectively.

Limitations

This work has several limitations that should be acknowledged:

1. Synthetic data generation relies on LLM inference pipelines (Gemma-4 via local/hosted endpoints), introducing computational resource and latency considerations for large-scale replication.
2. While we benchmarked two encoder-decoder architectures (mT5-small and IndicBART), decoder-only LLMs (e.g., LLaMA-3, Gemma-4) have not been evaluated for direct zero-shot or fine-tuned text normalization.
3. Evaluation is currently focused on intrinsic text normalization metrics (BLEU, chrF++, WER); downstream task evaluations (e.g., ASR post-processing, translation) remain for future work.
4. Synthetic augmentation exhibited a slight negative yield on Varhadi (-0.84 BLEU for mT5-small), suggesting that for already high-performing dialects, over-generalized suffix rules in synthetic data may slightly dilute precision—highlighting the need for targeted augmentation strategies.

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